Valkey

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What is Valkey?
Valkey is an open source (BSD) high-performance, in-memory, key/value datastore that supports a variety of workloads such as caching, session store, message queues, and can act as a primary database. Valkey can run as either a standalone daemon or in a cluster, with options for replication and high availability (valkey.io). Valkey was created to ensure the existence of a key/value datastore not under the control of any single vendor.

Why Valkey?
It is a high velocity, community driven, open source project. The Technical Steering Committee made up of members from Google, AWS, Percona, Ericsson, Tencent, and Alibaba. The project and roadmap is not centralized to any one company's priorities but is a collaborative effort . Valkey is production-ready and has partners like Percona providing enterprise support. In addition to self-hosting, it is available on clouds and platforms including Google Cloud, AWS, Oracle, Alibaba, Aiven, Anynines, Tencent and UpCloud. Participants include:

aiven aws Alibaba Cloud HUAWEI

BetterDB ByteDance DigitalOcean

ERICSSON Google Cloud HEROKU UpCloud

momento ORACLE PERCONA

Tencent Cloud verizon anynines

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Whats New?

Multi-threaded I/O

The latest versions of Valkey feature significant improvements to the existing I/O threading system, allowing the main thread and I/O threads to operate concurrently. This release also includes a number of improvements to offload work to the I/O threads an introduces efficient batching of commands.

Increased Memory Efficiency

Through the memory efficiency achieved by introducing dictionary per slot and key embedding into dictionary entry, users should have additional capacity to store more keys per node in Valkey. (up to 40%, but it will vary based on the workload).

Vector Similarity Search

Extends Valkey with vector search and secondary indexing capabilities. It enables users to index and query data stored in Valkey using complex queries with filters while maintaining high performance and scalability.

Valkey Bloom

Brings a native and space efficient probabilistic Module data type to Valkey. With this, users can create filters (space-efficient probabilistic Module data type) to add elements, perform “check” operation to test whether an element exists.

Full Text Search (FTS)

FTS in Valkey Search retrieves relevant documents by matching words and phrases within larger text fields when you don't have exact identifiers or structured values. It supports keyword matching, phrases, and pattern queries including prefix, suffix, wildcard, and fuzzy matching for typo tolerance.

Valkey JSON

Users can natively store, query, and modify JSON data structures using the JSONPath query language. The query expressions support advanced capabilities including wildcard selections, filter expressions, array slices, union operations, and recursive searches.

Where can you find us?

Valkey Website

GitHub Repository

Slack